



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)

Re-Accredited by NAAC with 'A++' Grade

Subject: Programming with Java

Name: Aman S. Choudhary

PRN: 25070122033

Div: CSE A1

Batch:2025-29

ASSIGNMENT No : 4

TITLE :Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Problem Statement

Develop a Java application demonstrating the use of wrapper classes for conversion between primitive data types and objects, and performing basic operations on wrapped values.

Learning Objectives

To perform conversion between primitive data types and wrapper objects using Java wrapper classes.

Theory

Wrapper classes are predefined classes that encapsulate primitive data types into objects. Java provides wrapper classes such as Integer, Double, Character, and Boolean for every primitive type. They support autoboxing, unboxing, parsing, and conversion methods. Wrapper classes are widely used in Java Collections Framework and when object representation of primitive values is required.

Outcome

By the end of this exercise, students will be able to convert between primitive data types and wrapper objects, and use wrapper class methods to perform validation and calculations on wrapped values.

Conclusion

Thus, we have successfully studied and implemented wrapper classes in Java. The Student Marks program demonstrated conversion of String values into Integer objects to calculate total marks, while the Employee Payroll System demonstrated how wrapper classes (Integer, Double) can be used to accept user input, validate salary values, and compute the net salary. The programs were executed successfully, and the outputs verified the correct working of wrapper class conversions.

Exercise

1. Create a program to convert student marks from String format to Integer and calculate total marks.

```
1 public class StudentMarks {  
    Run | Debug  
2     public static void main(String[] args) {  
3         String[] marksStr = {"85", "90", "78", "92", "88"};  
4         int total = 0;  
5         for (String m : marksStr) {  
6             Integer mark = Integer.parseInt(m); // String to Integer conversion  
7             total = total + mark;  
8         }  
9         System.out.println("Total Marks: " + total);  
10    }  
11 }
```

Total Marks: 433

2. Develop an **Employee Payroll System** that accepts employee IDs, basic salary, and bonus amounts from the user. Convert the entered values into wrapper objects and perform validation operations to ensure valid salary values before calculating the net salary.

```

1  import java.util.Scanner;
2
3  public class EmployeePayroll {
    Run | Debug
4      public static void main(String[] args) {
5          Scanner sc = new Scanner(System.in);
6
7          System.out.print(s: "Enter Employee ID: ");
8          Integer empId = Integer.parseInt(sc.nextLine());
9
10         System.out.print(s: "Enter Basic Salary: ");
11         Double basicSalary = Double.parseDouble(sc.nextLine());
12
13         System.out.print(s: "Enter Bonus Amount: ");
14         Double bonus = Double.parseDouble(sc.nextLine());
15
16         // Validation using wrapper values
17         if (basicSalary < 0 || bonus < 0) {
18             System.out.println(x: "Invalid salary or bonus values!");
19         } else {
20             double netSalary = basicSalary + bonus;
21             System.out.println("Employee ID: " + empId);
22             System.out.println("Net Salary: " + netSalary);
23         }
24
25         sc.close();
26     }
27 }

```

```

Enter Employee ID: 033
Enter Basic Salary: 395000
Enter Bonus Amount: 20000
Employee ID: 33
Net Salary: 415000.0

```
